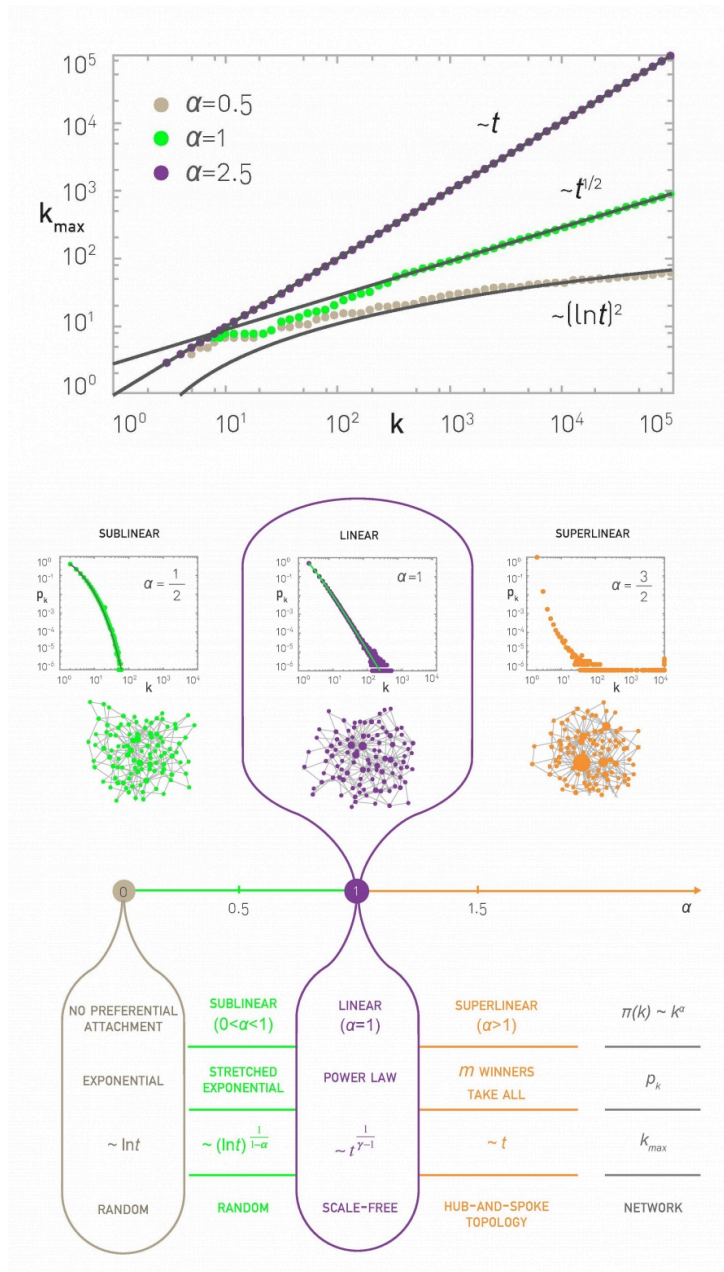


L12: Nonlinear Preferential Attachment



There are many variations of the PA model in the literature. Some of them generate special types of networks (such as directed or bipartite) while others include additional processes such as the removal or rewiring of edges or the aging of nodes.

Here, we present a nonlinear variation of the PA model in which the probability $\Pi(k)$ that a new node at time t connects to a specific node v that has degree- k is, not proportional to k , but proportional to:

$$\Pi(k) = c k^\alpha$$

where c is a constant (calculated so that $\sum_k p_k = 1$) and α is a positive exponent that may be larger or smaller than one. Of course the basic PA model results from $\alpha = 1$.

This nonlinear PA model can be analyzed mathematically using the same methodology we studied earlier (the rate-balance approach) – you can try it yourself or look at the textbook (Advanced Topics 5.C).

If $\alpha < 1$, the bias to connect to higher-degree nodes still exists but it is weaker than in the linear PA model. This changes the degree distribution qualitatively – it becomes the product of the power-law term $k^{-\alpha}$ and an exponential-decay term:

$$p_k \approx k^{-\alpha} e^{-m_\alpha k^{1-\alpha}}$$

where m_α is a constant that depends on α and m (the number of edges that each new node brings). This distribution is referred to as “*stretched exponential*” – the exponential term dominates for large values of k . The variance of the stretched exponential distribution does not diverge – such networks do not have large hubs and they do not exhibit the extreme properties of power-law networks we have seen in earlier lessons (such as the lack of an epidemic threshold).

Let’s also compare the degree dynamics of this model with the basic PA model we studied in the previous page. Recall that, when $\alpha = 1$, we saw that $k_{max} \approx \sqrt{t}$. When $0 < \alpha < 1$, we can show that k_{max} increases logarithmically,

$$k_{max} \approx (\ln t)^{1/(1-\alpha)}$$

What happens when $\alpha > 1$? Intuitively, the bias to connect to higher-degree nodes comes stronger. The resulting networks have fewer but larger hubs than $\alpha = 1$ – and the vast majority of the nodes connect only to those hubs. If α is sufficiently high all new nodes will connect to the first node because the degree of that node is higher, creating a hub-and-spoke (or “star topology”) network.

The maximum degree, in that case, increases linearly with “time”, $k_{max} \approx t$, because all new edges connect to the same node.

The previous results are illustrated with numerical results in the visualizations at the top of the page, showing the degree distribution for three values of α (0.5, 1 and 1.5 with $N=10,000$ nodes) and the maximum degree dynamics for $\alpha = 0.5, 1$ and 2.5.

Food For Thought

What happens when α tends to 0? What is the resulting degree distribution and how does k_{max} increase with time?